

Quantities, Units and Modelling

Mechanics is a model of the real world. Know the units, know which quantities have a direction, and know what each modelling word lets you assume.



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THE TOPIC ON ONE MAP

Quantities, Units and Modelling

01 SI units

kg, m, s, N $\text{km h}^{-1} \div 3.6$ $\text{g} \rightarrow \text{kg}$

02 Scalars and vectors

distance / displacement speed / velocity signs

03 Modelling words

particle light, inextensible smooth, rough, uniform

04 Weight and accuracy

$W = mg$ $g = 9.8: 2 \text{ or } 3 \text{ s.f.}$

THE FOUR BASIC SI UNITS OF M1

mass: kg length: m time: s force: $N = kg \, m \, s^{-2}$
 change everything to these before you start $km \, h^{-1} \div 3.6 = m \, s^{-1}$

01 SI Units and Conversions

Every formula in M1 works only in SI units. Before you write an equation, change grams to kg, cm to m, km to m, tonnes to kg, minutes and hours to seconds, and $km \, h^{-1}$ to $m \, s^{-1}$.

WHEN YOU SEE → YOU DO

"54 $km \, h^{-1}$ "	→ $\div 3.6$: $15 \, m \, s^{-1}$
"mass 250 g", "2 tonnes"	→ 0.25 kg, 2000 kg
"12 cm", "4.8 km"	→ 0.12 m, 4800 m
"3 minutes"	→ 180 s
"stating the units of your answer"	→ impulse $N \, s$ · moment $N \, m$ · speed $m \, s^{-1}$

quantity	unit	formula
velocity	$m \, s^{-1}$	$\frac{\text{displacement}}{\text{time}}$
acceleration	$m \, s^{-2}$	$\frac{\text{change of velocity}}{\text{time}}$
force, weight	N	$F = ma$, $W = mg$
momentum, impulse	$N \, s$	mv , Ft
moment	$N \, m$	$F \times d$

IDEA 01

Practise: SI Units and Conversions

Questions written to practise exactly this idea.

Q1 DR ESLAM Speed units

M1 · TOPIC 1 2 marks

Convert (a) 72 km h^{-1} to m s^{-1} (b) 15 m s^{-1} to km h^{-1} .

$\text{km h}^{-1} \div 3.6 = \text{m s}^{-1}$.

Q2 DR ESLAM Mass and length

M1 · TOPIC 1 2 marks

Write in SI units: (a) 450 g (b) 2.4 tonnes (c) 35 cm (d) 1.2 km.

Q3 DR ESLAM Units of a quantity

M1 · TOPIC 1 2 marks

State the SI units of (a) acceleration (b) force (c) momentum (d) moment of a force.

02 Scalars and Vectors

A scalar has only a size (distance, speed, mass, time). A vector has a size **and** a direction (displacement, velocity, acceleration, force, momentum). In a straight line the direction is just a sign: choose a positive direction and keep it.

WHEN YOU SEE → YOU DO

“speed” vs “velocity”

→ speed is the size of the velocity (never negative)

“distance” vs “displacement”

→ distance adds every part; displacement is start to finish

“rebounds”, “reverses direction”

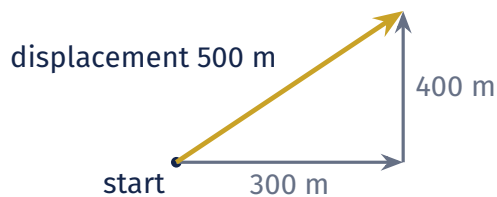
→ the velocity changes sign

“average speed”

→ total distance ÷ total time

a vector in i, j

→ size by Pythagoras (Topic 2)



distance = 700 m (scalar)

displacement = 500 m on 037° (vector)

IDEA 02

Practise: Scalars and Vectors

Questions written to practise exactly this idea.

Q4 DR ESLAM Scalar or vector?

M1 · TOPIC 1 2 marks

Sort into scalars and vectors: distance, displacement, speed, velocity, mass, weight, time, acceleration.

Q5 DR ESLAM Distance vs displacement

M1 · TOPIC 1 3 marks

A girl walks 300 m east then 400 m north in 500 s. Find her distance, her displacement (size and bearing), her average speed and the size of her average velocity.

Q6 DR ESLAM Velocity with a sign

M1 · TOPIC 1 2 marks

A ball moves up at 6 m s^{-1} and later down at 4 m s^{-1} . Taking up as positive, write both velocities and the change in velocity.

03 Modelling Words and What They Mean

The exam often asks: “state how you used the fact that ...”. Each modelling word gives you one thing in the working. Learn the pairs below — they are free marks.

WHEN YOU SEE → YOU DO

“modelled as a particle”

→ all the mass at one point; no size, no turning, no air resistance

“light string / rod”

→ no mass: the tension is the same all along it

“inextensible string”

→ both ends move with the same speed and acceleration

“smooth pulley” / “smooth surface”

→ no friction: same tension on both sides / $F = 0$

“rough surface”

→ friction acts: $F \leq \mu R$

“uniform rod”

→ weight acts at the midpoint

“rigid rod” / “beam stays straight”

→ it does not bend, so distances stay fixed

“state one way the model could be improved”

→ air resistance, size of the object, resistance varying with speed

ZONE THE BODY

particle, uniform, rigid

where the weight acts;
no size

ZONE THE STRING

light, inextensible

same T , same a

ZONE THE SURFACE

smooth or rough

$F = 0$ or $F \leq \mu R$

IDEA 03

Practise: Modelling Words and What They Mean

Questions written to practise exactly this idea.

Q7 **DR ESLAM** What does it mean?

M1 · TOPIC 1 4 marks

State what each modelling word means for your working: (a) particle (b) light string (c) inextensible string (d) smooth pulley (e) uniform rod (f) rough surface.

Q8 **DR ESLAM** Improve the model

M1 · TOPIC 1 2 marks

A car is modelled as a particle moving against a constant resistance. State two ways the model could be made more realistic.

04 Weight, g and Accuracy

Weight is a force: $W = mg$, acting straight down, in newtons. Mass (kg) never changes; weight depends on g . Take $g = 9.8 \text{ m s}^{-2}$ unless told otherwise, and because 9.8 has 2 significant figures, give final answers to 2 or 3 s.f.

WHEN YOU SEE → YOU DO

“a mass of 3 kg”

→ weight $3g = 29.4 \text{ N}$

“in terms of g ”

→ leave g as a letter

“give your answer to an appropriate degree of accuracy”

→ 2 or 3 s.f. when $g = 9.8$ was used

an exact form is asked (“show that $a = \frac{g}{5}$ ”)

→ keep g and fractions — no decimals

DR ESLAM’S TIP Keep full calculator values until the last line, then round once. Rounding in the middle is the most common way to lose the final accuracy mark.

IDEA 04

Practise: Weight, g and Accuracy

Questions written to practise exactly this idea.

Q9 DR ESLAM Weight and mass

M1 · TOPIC 1 2 marks

Find the weight of a 65 kg person. What is their mass on the Moon, where $g = 1.6 \text{ m s}^{-2}$, and their weight there?

Q10 DR ESLAM How many figures?

M1 · TOPIC 1 2 marks

A calculation using $g = 9.8$ gives $T = 23.5198 \text{ N}$. How should you give the answer, and why?

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-------------------------|---|
| 01 | SI units | kg, m, s, N; $\text{km h}^{-1} \div 3.6$. |
| 02 | Scalars, vectors | Vectors have direction: choose + and keep it. |
| 03 | Modelling words | Particle, light, inextensible, smooth, rough, uniform. |
| 04 | Weight, accuracy | $W = mg$; $g = 9.8 \rightarrow 2 \text{ or } 3 \text{ s.f.}$ |

BOTTOM LINE SI units first • a sign for every direction • each modelling word gives one fact.

Answers

Quantities, Units and Modelling • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SI UNITS AND CONVERSIONS

- Q1** Dr Eslam • Speed units (a) 20 m s^{-1} (b) 54 km h^{-1}
- Q2** Dr Eslam • Mass and length (a) 0.45 kg (b) 2400 kg (c) 0.35 m (d) 1200 m
- Q3** Dr Eslam • Units of a quantity (a) m s^{-2} (b) $\text{N} (= \text{kg m s}^{-2})$ (c) $\text{N s} (= \text{kg m s}^{-1})$ (d) N m

IDEA 02 • SCALARS AND VECTORS

- Q4** Dr Eslam • Scalar or vector? scalars: distance, speed, mass, time; vectors: displacement, velocity, weight, acceleration
700 m; 500 m on 037° ; 1.4 m s^{-1} ; 1 m s^{-1}
- Q5** Dr Eslam • Distance vs displacement
- Q6** Dr Eslam • Velocity with a sign +6, -4; change = -10 m s^{-1}

IDEA 03 • MODELLING WORDS AND WHAT THEY MEAN

- Q7** Dr Eslam • What does it mean? (a) mass at one point, no size or rotation (b) no mass: same tension along it (c) same speed and acceleration at both ends (d) no friction: same tension on both sides (e) weight at the midpoint (f) friction acts, $F \leq \mu R$
- Q8** Dr Eslam • Improve the model e.g. resistance increasing with speed; include air resistance; car has size (not a particle); driving force not constant

IDEA 04 • WEIGHT, g AND ACCURACY

- Q9** Dr Eslam · Weight and mass 637 N; mass still 65 kg; weight 104 N
- Q10** Dr Eslam · How many figures? 23.5 N (3 s.f.) or 24 N (2 s.f.) — $g = 9.8$ is given to 2 s.f.